Tenno: Data Privacy Act

Falcon-7B QLoRA Fine-Tuning Notebook

Objective

This notebook performs parameter-efficient fine-tuning (PEFT) on Falcon-7B using QLoRA (4-bit quantization + LoRA adapters). Training samples are constructed from the frozen baseline prompt set by retrieving relevant legal context (FAISS) and using the expected outputs as supervised targets. After fine-tuning, the model is evaluated on the same prompts under the same retrieval setup and outputs are saved for comparison in the main notebook.

Section 1: Enviroment Setup

1.1 Install Required Packages

```
!pip uninstall -y torch torchvision torchaudio
!pip cache purge
!pip install -q --upgrade torch torchvision torchaudio --index-url
https://download.pytorch.org/whl/cu121
!pip install -q faiss-cpu
!pip install -q -U "bitsandbytes>=0.46.1"
```

```
Found existing installation: torch 2.10.0+cu128
Uninstalling torch-2.10.0+cu128:
  Successfully uninstalled torch-2.10.0+cu128
Found existing installation: torchvision 0.25.0+cu128
Uninstalling torchvision-0.25.0+cu128:
  Successfully uninstalled torchvision-0.25.0+cu128
Found existing installation: torchaudio 2.10.0+cu128
Uninstalling torchaudio-2.10.0+cu128:
  Successfully uninstalled torchaudio-2.10.0+cu128
WARNING: No matching packages
Files removed: 0
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0:00:00
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```

```
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0:00:00
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0:00:00
```

Installs the PEFT fine-tuning stack (LoRA), quantization (bitsandbytes), dataset utilities, retrieval tooling (FAISS), and sentence embeddings.

1.2 Mount Google Drive

```
from google.colab import drive
drive.mount("/content/drive")
```

```
Mounted at /content/drive
```

Mounts Google Drive so we can load FAISS artifacts, prompts, and store trained adapters + outputs.

1.3 Persistent Hugging Face Cache (Drive)

```
import os

PROJECT_DIR = "/content/drive/MyDrive/Mapua /3rd Year Term 2/CSS181-
3/NLP_Project"
HF_CACHE_DIR = f"{PROJECT_DIR}/Models/hf_cache"
os.makedirs(HF_CACHE_DIR, exist_ok=True)

os.environ["HF_HOME"] = HF_CACHE_DIR
os.environ["HUGGINGFACE_HUB_CACHE"] = HF_CACHE_DIR
os.environ["TRANSFORMERS_CACHE"] = HF_CACHE_DIR

print("□ Using persistent HF cache:", HF_CACHE_DIR)
```

```
□ Using persistent HF cache: /content/drive/MyDrive/Mapua /3rd Year
Term 2/CSS181-3/NLP_Project/Models/hf_cache
```

Ensures model downloads persist across sessions.

1.4 Define Paths + Load Prompts and FAISS Artifacts

```
import json
import faiss

PROMPTS_FILE = f"{PROJECT_DIR}/Prompts/baseline_prompts_v1.json"
FAISS_INDEX_PATH = f"{PROJECT_DIR}/Chunks/faiss_index.bin"
FAISS_META_PATH  = f"{PROJECT_DIR}/Chunks/faiss_metadata.json"

RESULTS_DIR = f"{PROJECT_DIR}/Results"
os.makedirs(RESULTS_DIR, exist_ok=True)

ADAPTER_DIR = f"{PROJECT_DIR}/Models/falcon_finetune_adapter"
os.makedirs(ADAPTER_DIR, exist_ok=True)

with open(PROMPTS_FILE, "r", encoding="utf-8") as f:
    prompts = json.load(f)

index = faiss.read_index(FAISS_INDEX_PATH)
with open(FAISS_META_PATH, "r", encoding="utf-8") as f:
    meta = json.load(f)

print("□ Prompts:", len(prompts))
print("□ FAISS vectors:", index.ntotal)
print("□ Meta rows:", len(meta))
print("□ Adapter output dir:", ADAPTER_DIR)
```

```
□ Prompts: 10
□ FAISS vectors: 198
□ Meta rows: 198
```

```
 Adapter output dir: /content/drive/MyDrive/Mapua /3rd Year Term
2/CSS181-3/NLP_Project/Models/falcon_finetune_adapter
```

Loads frozen prompts and retrieval artifacts.

1.4.1 Artifact Safety and Versioning

```
import os, shutil, time

def safe_copy_file(src, dst):
    if os.path.exists(src):
        shutil.copy(src, dst)
        print(f" Copied: {src} -> {dst}")
    else:
        print(f"⚠ Not found (skip): {src}")

def safe_move_dir(src, dst):
    if os.path.exists(src):
        if os.path.exists(dst):
            shutil.rmtree(dst)
            print(f" Removed existing: {dst}")
        shutil.move(src, dst)
        print(f" Moved: {src} -> {dst}")
    else:
        print(f"⚠ Not found (skip): {src}")

# Canonical/original filenames
PHASE1_JSON_CANON = f"{RESULTS_DIR}/finetuned_outputs_falcon.json"
PHASE1_ADAPTER_CANON = f"{PROJECT_DIR}/Models/falcon_finetune_adapter"

# Frozen copy names
PHASE1_JSON_ORIGINAL =
f"{RESULTS_DIR}/finetuned_outputs_falcon_phase1_original.json"
PHASE1_ADAPTER_ORIGINAL =
f"{PROJECT_DIR}/Models/falcon_finetune_adapter_phase1_original"

# Rerun names
PHASE1_JSON_RERUN =
f"{RESULTS_DIR}/finetuned_outputs_falcon_phase1_rerun.json"
PHASE1_ADAPTER_RERUN =
f"{PROJECT_DIR}/Models/falcon_finetune_adapter_phase1_rerun"

print(" Version paths prepared.")
```

```
 Version paths prepared.
```

1.5 Reproducibility Setup

```
import random
import numpy as np
import torch

seed = 42

random.seed(seed)
np.random.seed(seed)
torch.manual_seed(seed)
torch.cuda.manual_seed_all(seed)

torch.backends.cudnn.deterministic = True
torch.backends.cudnn.benchmark = False
```

A fixed seed is set at the start of the notebook to improve reproducibility across repeated runs of dataset construction, model initialization, and fine-tuning.

1.6 Load SentenceTransformer Embedde

```
import numpy as np
from sentence_transformers import SentenceTransformer

embed_model_name = "all-MiniLM-L6-v2"
embedder = SentenceTransformer(embed_model_name)
print("□ Loaded embedder:", embed_model_name)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/
_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your
settings tab (https://huggingface.co/settings/tokens), set it as
secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to
access public models or datasets.
  warnings.warn(
```

{"model_id":"269d05be733b45b391bc6f644f00110b","version_major":2,"version_minor":0}

```
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                      | Status     |  |
-------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different
task/architecture; not ok if you expect identical arch.
```

```
Warning: You are sending unauthenticated requests to the HF Hub.
Please set a HF_TOKEN to enable higher rate limits and faster
downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending
unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
enable higher rate limits and faster downloads.

 Loaded embedder: all-MiniLM-L6-v2
```

Uses the same embedder used to build the FAISS index.

Section 2: Shared Retrieval & Prompt Construction Utilities

2.1 Retrieval Helper + Prompt Builder

```
TOP_K = 3

def retrieve_top_k_chunks(query: str, top_k: int = TOP_K):
    q_emb = embedder.encode([query],
convert_to_numpy=True).astype(np.float32)
    distances, indices = index.search(q_emb, top_k)
    idxs = indices[0].tolist()

    retrieved = []
    for rank, i in enumerate(idxs):
        row = meta[i]
        retrieved.append({
            "faiss_id": i,
            "text": row.get("text", ""),
            "source": row.get("source", ""),
            "page_number": row.get("page_number", None),
            "distance": float(distances[0][rank]),
        })
    return retrieved

def build_rag_input(retrieved_chunks, question: str) -> str:
    context_blocks = []
    for c in retrieved_chunks:
        src = c["source"]
        pg = c["page_number"]
        txt = c["text"]
        context_blocks.append(f"[Source: {src} | Page: {pg}]\n{txt}")
```

```
    context = "\n\n---\n\n".join(context_blocks)

    return (
        "You are a legal assistant. Answer the question using ONLY the
retrieved context.\n"
        "If the context does not contain the answer, say: \"Not found
in the provided context.\"\n\n"
        f"Retrieved Context (Top-{TOP_K}):\n{context}\n\n"
        f"Question:\n{question}\n\n"
        "Answer:"
    )

def join_expected_output(exp):
    if exp is None:
        return ""
    if isinstance(exp, list):
        return "\n".join([str(x).strip() for x in exp if
str(x).strip()])
    return str(exp).strip()
```

Defines the same retrieval + RAG formatting used in baseline, ensuring fair before/after comparison.

Section 3: Phase 1 - Initial Fine-Tuning (TOP_K = 3)

3.1 Dataset Construction (TOP_K = 3)

```
from datasets import Dataset

train_rows = []
for p in prompts:
    question = p.get("question", "")
    expected = join_expected_output(p.get("expected_output"))

    # Build training input with retrieved context
    retrieved = retrieve_top_k_chunks(question, top_k=TOP_K)
    prompt_text = build_rag_input(retrieved, question)

    # Supervised target (expected answer)
    # We train the model to produce the expected output given the same
RAG prompt.
    train_rows.append({
```

```
        "prompt_id": p.get("prompt_id"),
        "prompt_text": prompt_text,
        "target_text": expected
    })

ds_train = Dataset.from_list(train_rows)
print("□ Training samples:", len(ds_train))
print(ds_train[0]["prompt_text"][:400])
print("\n--- TARGET ---\n", ds_train[0]["target_text"])
```

```
□ Training samples: 10
You are a legal assistant. Answer the question using ONLY the
retrieved context.
If the context does not contain the answer, say: "Not found in the
provided context."

Retrieved Context (Top-3):
[Source: RA-10173-Data-Privacy-Act-of-2012.pdf | Page: 6]
authorizing their processing. the personal information controller must
ensure implementation of personal information processing principles
set out

--- TARGET ---
 Consent of the data subject
Fulfilling contractual obligations
Compliance with legal obligations
Protection of vitally important interests
Responding to national emergency or public order
Fulfillment of statutory mandate of a public authority
Legitimate interests of controller or third parties, except when
overridden by fundamental rights
```

Creates supervised fine-tuning samples using your existing prompts + expected outputs. This is a small dataset, but valid for demonstrating PEFT.

3.2 Model Initialization (4-bit QLoRA)

```
import gc
import torch
from transformers import AutoTokenizer, AutoModelForCausalLM,
BitsAndBytesConfig

# ⚠ Set this to the SAME Falcon model you used in baseline
MODEL_ID = "tiiuae/falcon-7b-instruct"

bnb_config = BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_compute_dtype=torch.float16,
```

```
    bnb_4bit_use_double_quant=True,
    bnb_4bit_quant_type="nf4",
)

print("Loading Falcon tokenizer...")
tokenizer = AutoTokenizer.from_pretrained(MODEL_ID,
cache_dir=HF_CACHE_DIR, use_fast=True)

if tokenizer.pad_token is None:
    tokenizer.pad_token = tokenizer.eos_token

gc.collect()
torch.cuda.empty_cache()
os.environ["PYTORCH_CUDA_ALLOC_CONF"] = "expandable_segments:True"

print("Loading Falcon model (4-bit)...")
model = AutoModelForCausalLM.from_pretrained(
    MODEL_ID,
    cache_dir=HF_CACHE_DIR,
    device_map={"": 0},
    quantization_config=bnb_config,
    low_cpu_mem_usage=True,
    force_download=True
)
model.train()
print("□ Falcon loaded for fine-tuning.")
```

```
Loading Falcon tokenizer...
Loading Falcon model (4-bit)...
```


```
{"model_id":"e6bd1b41871543e3907f32638ec3499b","version_major":2,"vers
ion_minor":0}

{"model_id":"ba046c71bca74721bf1055a36420d604","version_major":2,"vers
ion_minor":0}

{"model_id":"5fb1d8f09f8c4aef90a22e0989f2c2e5","version_major":2,"vers
ion_minor":0}

{"model_id":"5747a1c7349443f79f88f85dded69709","version_major":2,"vers
ion_minor":0}

{"model_id":"e21926fc3444467da06e8c748e87044d","version_major":2,"vers
ion_minor":0}

{"model_id":"a47e320190224915a297e2e7bdeaa2e2","version_major":2,"vers
ion_minor":0}

{"model_id":"1fd1702191c64b0d89501bea2128d100","version_major":2,"vers
ion_minor":0}
```

```
The tied weights mapping and config for this model specifies to tie
transformer.word_embeddings.weight to lm_head.weight, but both are
present in the checkpoints, so we will NOT tie them. You should update
the config with `tie_word_embeddings=False` to silence this warning
```

{"model_id":"2addb84fc5c740a485d7d74cb2add219","version_major":2,"version_minor":0}

```
 Falcon loaded for fine-tuning.
```

Loads Falcon in 4-bit for QLoRA fine-tuning.

3.3 LoRA Configuration

```
from peft import LoraConfig, get_peft_model,
prepare_model_for_kbit_training

model = prepare_model_for_kbit_training(model)

lora_config = LoraConfig(
    r=16,
    lora_alpha=32,
    lora_dropout=0.05,
    bias="none",
    task_type="CAUSAL_LM",
)

model = get_peft_model(model, lora_config)
model.print_trainable_parameters()
```

```
trainable params: 4,718,592 || all params: 7,221,908,352 || trainable
%: 0.0653
```

Enables PEFT adapters so you train only a small number of parameters.

3.4 Tokenization & Loss Masking

```
MAX_LEN = 1024

def tokenize_and_mask(batch):
    prompt = batch["prompt_text"]
    target = batch["target_text"]

    full = prompt + " " + target
    tok_full = tokenizer(full, truncation=True, max_length=MAX_LEN)

    tok_prompt = tokenizer(prompt, truncation=True,
```

```
max_length=MAX_LEN)
    prompt_len = len(tok_prompt["input_ids"])

    labels = tok_full["input_ids"].copy()
    # mask prompt tokens so loss is computed only on target_text
    labels[:prompt_len] = [-100] * prompt_len

    tok_full["labels"] = labels
    return tok_full

ds_train_tok = ds_train.map(tokenize_and_mask,
remove_columns=ds_train.column_names)
```

{"model_id":"075433e45f1b4cb5835623ed66c8d1bc","version_major":2,"version_minor":0}

This trains the model to generate the expected answer, not to “memorize” the prompt text.

3.5 Training Run

```
from transformers import TrainingArguments, Trainer,
DataCollatorForLanguageModeling

training_args = TrainingArguments(
    output_dir=f"{PROJECT_DIR}/Models/falcon_finetune_ckpts",
    per_device_train_batch_size=1,
    gradient_accumulation_steps=8,
    num_train_epochs=5,
    learning_rate=2e-4,
    fp16=True,
    logging_steps=1,
    save_strategy="epoch",
    report_to="none"
)

data_collator = DataCollatorForLanguageModeling(tokenizer=tokenizer,
mlm=False)

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=ds_train_tok,
    data_collator=data_collator
)

trainer.train()
```

```
/usr/local/lib/python3.12/dist-packages/torch/_dynamo/
eval_frame.py:632: UserWarning: torch.utils.checkpoint: the
use_reentrant parameter should be passed explicitly. In version 2.5 we
will raise an exception if use_reentrant is not passed.
use_reentrant=False is recommended, but if you need to preserve the
current default behavior, you can pass use_reentrant=True. Refer to
docs for more details on the differences between the two variants.
  return fn(*args, **kwargs)
```

```
<IPython.core.display.HTML object>
```

```
/usr/local/lib/python3.12/dist-packages/torch/_dynamo/
eval_frame.py:632: UserWarning: torch.utils.checkpoint: the
use_reentrant parameter should be passed explicitly. In version 2.5 we
will raise an exception if use_reentrant is not passed.
use_reentrant=False is recommended, but if you need to preserve the
current default behavior, you can pass use_reentrant=True. Refer to
docs for more details on the differences between the two variants.
  return fn(*args, **kwargs)
/usr/local/lib/python3.12/dist-packages/torch/_dynamo/eval_frame.py:63
2: UserWarning: torch.utils.checkpoint: the use_reentrant parameter
should be passed explicitly. In version 2.5 we will raise an exception
if use_reentrant is not passed. use_reentrant=False is recommended,
but if you need to preserve the current default behavior, you can pass
use_reentrant=True. Refer to docs for more details on the differences
between the two variants.
  return fn(*args, **kwargs)
/usr/local/lib/python3.12/dist-packages/torch/_dynamo/eval_frame.py:63
2: UserWarning: torch.utils.checkpoint: the use_reentrant parameter
should be passed explicitly. In version 2.5 we will raise an exception
if use_reentrant is not passed. use_reentrant=False is recommended,
but if you need to preserve the current default behavior, you can pass
use_reentrant=True. Refer to docs for more details on the differences
between the two variants.
  return fn(*args, **kwargs)
/usr/local/lib/python3.12/dist-packages/torch/_dynamo/eval_frame.py:63
2: UserWarning: torch.utils.checkpoint: the use_reentrant parameter
should be passed explicitly. In version 2.5 we will raise an exception
if use_reentrant is not passed. use_reentrant=False is recommended,
but if you need to preserve the current default behavior, you can pass
use_reentrant=True. Refer to docs for more details on the differences
between the two variants.
  return fn(*args, **kwargs)
```

```
TrainOutput(global_step=10, training_loss=1.9190805435180665,
metrics={'train_runtime': 454.5094, 'train_samples_per_second': 0.11,
'train_steps_per_second': 0.022, 'total_flos': 2071697993433600.0,
'train_loss': 1.9190805435180665, 'epoch': 5.0})
```

Runs PEFT fine-tuning. With only 10 samples, this is fast and mainly helps demonstrate adapter-based tuning.

3.6 Adapter Export

```
model.save_pretrained(ADAPTER_DIR)
tokenizer.save_pretrained(ADAPTER_DIR)
print("□ Saved LoRA adapter + tokenizer to:", ADAPTER_DIR)
```

```
□ Saved LoRA adapter + tokenizer to: /content/drive/MyDrive/Mapua /3rd Year Term 2/CSS181-3/NLP_Project/Models/falcon_finetune_adapter
```

Saves LoRA weights (small) so you can reload without re-training.

3.6.1 Artifact Version Control

To maintain reproducibility and prevent unintended overwriting of previous experimental artifacts, Phase 1 outputs are versioned prior to proceeding with additional training runs. Existing JSON results are preserved, and the exported adapter is moved to a version-specific directory. This ensures clean separation between experimental iterations.

```
# Preserve existing JSON results (if present)
if os.path.exists(PHASE1_JSON_CANON):
    shutil.copy(PHASE1_JSON_CANON, PHASE1_JSON_ORIGINAL)
    print("□ Existing Phase 1 JSON preserved.")
else:
    print("⚠ No existing Phase 1 JSON found to preserve.")

# Move current adapter to versioned folder
if os.path.exists(PHASE1_ADAPTER_CANON):
    if os.path.exists(PHASE1_ADAPTER_RERUN):
        shutil.rmtree(PHASE1_ADAPTER_RERUN)
    shutil.move(PHASE1_ADAPTER_CANON, PHASE1_ADAPTER_RERUN)
    print("□ Phase 1 adapter moved to versioned directory.")
else:
    print("⚠ Phase 1 adapter directory not found.")
```

```
□ Existing Phase 1 JSON preserved.
□ Phase 1 adapter moved to versioned directory.
```

3.7 Evaluation Model Setup

```
import gc
import torch

# Switch trained model to eval mode
```

```
model.eval()

# Free any cached memory
gc.collect()
torch.cuda.empty_cache()

ft_model = model  # use the already-loaded fine-tuned model
print("□ Using in-memory fine-tuned model for evaluation (no
reload).")
```

```
□ Using in-memory fine-tuned model for evaluation (no reload).
```

Loads fine-tuned adapters on top of the base Falcon model.

3.8 Deterministic Generation

```
@torch.inference_mode()
def generate_answer_ft(input_text: str, max_new_tokens: int = 256) ->
str:
    inputs = tokenizer(input_text,
return_tensors="pt").to(ft_model.device)

    out_ids = ft_model.generate(
        **inputs,
        do_sample=False,
        temperature=0.0,
        top_p=1.0,
        max_new_tokens=max_new_tokens,
        pad_token_id=tokenizer.eos_token_id
    )

    decoded = tokenizer.decode(out_ids[0], skip_special_tokens=True)
    return decoded[len(input_text):].strip()
```

Keeps evaluation deterministic to compare fairly against baseline.

3.9 Evaluation & JSON Export

```
from tqdm import tqdm

FT_OUTPUT_PATH = PHASE1_JSON_RERUN

ft_results = []

for p in tqdm(prompts, desc="Running Falcon Fine-Tuned RAG prompts"):
    q = p["question"]
```

```
    retrieved = retrieve_top_k_chunks(q, top_k=TOP_K)
    input_text = build_rag_input(retrieved, q)
    answer = generate_answer_ft(input_text, max_new_tokens=256)

    ft_results.append({
        "model_id": MODEL_ID,
        "adapter_dir": ADAPTER_DIR,
        "prompt_id": p.get("prompt_id"),
        "method": p.get("method"),
        "condition": p.get("condition"),
        "dataset_file": p.get("dataset_file"),
        "section_index": p.get("section_index"),
        "question": q,
        "retrieved_chunks": retrieved,
        "answer": answer,
        "expected_output": p.get("expected_output")
    })

with open(FT_OUTPUT_PATH, "w", encoding="utf-8") as f:
    json.dump(ft_results, f, indent=2, ensure_ascii=False)

print("□ Saved fine-tuned outputs to:", FT_OUTPUT_PATH)
```

```
Running Falcon Fine-Tuned RAG prompts:   0%|          | 0/10 [00:00<?,
?it/s]The following generation flags are not valid and may be ignored:
['temperature']. Set `TRANSFORMERS_VERBOSITY=info` for more details.
Running Falcon Fine-Tuned RAG prompts: 100%|██████████| 10/10
[02:18<00:00, 13.87s/it]

□ Saved fine-tuned outputs to: /content/drive/MyDrive/Mapua /3rd Year
Term
2/CSS181-3/NLP_Project/Results/finetuned_outputs_falcon_phase1_rerun.j
son
```

Creates the “fine-tuned Falcon outputs” file that your Original Notebook will load in Section 6/7.

3.10 Output Preview

```
print(f"\n□ Loaded {len(ft_results)} fine-tuned outputs\n")

for item in ft_results:
    print("=" * 80)
    print(f"[Prompt {item['prompt_id']}] ({item['method']})")
    print("-" * 80)
    print("Q:", item["question"])
```

```
    print("\nA:", item["answer"][:400])
    print("\n")


 Loaded 10 fine-tuned outputs

==================================================================
==========
[Prompt 1] (Zero-Shot)
------------------------------------------------------------------
----------
Q: List all the conditions under which the processing of personal
information is considered lawful according to this section.

A: The processing of personal information is considered lawful
according to this section when:

1. The personal information is collected from the data subject.
2. The personal information is collected for a specific purpose.
3. The personal information is collected from a reliable source.
4. The personal information is collected for a specific purpose.
5. The personal information is collected from a


==================================================================
==========
[Prompt 2] (Zero-Shot)
------------------------------------------------------------------
----------
Q: What are the penalties for unauthorized disclosure of personal
information and sensitive personal information under this section?

A: The penalties for unauthorized disclosure of personal information
and sensitive personal information under this section are imprisonment
ranging from one (1) year to three (3) years and a fine of not less
than five hundred thousand pesos (php500,000.00) but not more than one
million pesos (php1,000,000.00).


==================================================================
==========
[Prompt 3] (Chain-of-Thought Prompting)
------------------------------------------------------------------
----------
Q: Based on the legal definition, list what qualifies as sensitive
personal information and briefly explain why each qualifies.

A: 1. Race, ethnic origin, marital status, age, color, and religious,
philosophical or political affiliations: These are personal
information that are considered sensitive because they are often used
```

to identify a person and may be used to discriminate against them.

2. Health, education, genetic or sexual life of an individual: These are personal information that are considered sensitive because the

==========
[Prompt 4] (Chain-of-Thought Prompting)

Q: Explain the general data privacy principles outlined in this section and why each principle is important.

A: The general data privacy principles outlined in this section are:

1. Data security: It is important to ensure that personal data is kept secure and protected from unauthorized access, use, or disclosure. This is crucial to ensure that data privacy is maintained and that data is not compromised.

2. Data accuracy: It is important to ensure that personal data is accurate and up-to-date. This is imp

==========
[Prompt 5] (One-Shot Learning Prompt)

Q: List the rights granted to data subjects under this section.

A: The rights granted to data subjects under this section are:

1. Right to access personal data
2. Right to data portability
3. Right to be informed
4. Right to rectification
5. Right to erasure
6. Right to restriction of processing
7. Right to object to processing
8. Right to data portability
9. Right to withdraw consent
10. Right to be forgotten
11. Right to be informed of a data breach
12. Right

==========

[Prompt 6] (One-Shot Learning Prompt)

Q: Summarize the breach notification requirement.

A: The breach notification requirement is that a personal information controller must notify the National Privacy Commission (NPC) of any security breach or personal data breach that may have occurred. The notification must include the type of breach, the personal data that was affected, the date of the breach, and the contact information of the person responsible for the breach. The notification mus

==========
[Prompt 7] (One-Shot Learning Prompt)

Q: Explain what the principle of proportionality means.

A: The principle of proportionality means that the processing of personal data should be in proportion to the purpose for which it is being collected, processed, and retained. In other words, the processing of personal data should be in proportion to the purpose of the data being collected, processed, and retained. This means that the processing of personal data should be in proportion to the purpose

==========
[Prompt 8] (One-Shot Learning Prompt)

Q: List the three types of security measures required.

A: 1. Physical security measures
2. Data security measures
3. Network security measures

==========
[Prompt 9] (Negative Prompting)

Q: List only the lawful processing conditions mentioned.

A: 1. Consent of the data subject
2. Processing is necessary for the performance of a contract
3. Processing is necessary for the purpose of a legitimate interest

```
pursued by the controller or a third party
4. Processing is necessary for the purpose of a public interest
5. Processing is necessary for the purpose of a public authority
6. Processing is necessary for the purpose of a public interest
7. P


==================================================================
==========
[Prompt 10] (Negative Prompting)
------------------------------------------------------------------
----------
Q: Identify the penalties for unauthorized disclosure.

A: The penalties for unauthorized disclosure are imprisonment ranging
from one (1) year to three (3) years and a fine of not less than five
hundred thousand pesos (php500,000.00) but not more than one million
pesos (php1,000,000.00).
```

Section 4: Phase 2 - Refined Fine-Tuning (TOP_K = 5)

4.1 Rationale for Refinement

Phase 1 demonstrated a complete parameter-efficient fine-tuning pipeline but produced only marginal quantitative changes relative to the baseline model. A likely limitation is retrieval coverage. With TOP_K = 3, certain prompts may not include sufficient legal evidence in the retrieved context to fully support the expected outputs. This creates supervision mismatch, as the model is instructed to rely solely on retrieved context while being trained against complete reference answers.

To address this, Phase 2 increases retrieval coverage to TOP_K = 5 and slightly stabilizes training hyperparameters (reduced epochs and learning rate). This refinement aims to improve semantic alignment while maintaining strong contextual grounding.

4.2 Configuration

```
TOP_K_V2 = 5

ADAPTER_DIR_V2 =
```

```
f"{PROJECT_DIR}/Models/falcon_finetune_adapter_v2_topk5"
os.makedirs(ADAPTER_DIR_V2, exist_ok=True)

FT_OUTPUT_PATH_V2 =
f"{RESULTS_DIR}/finetuned_outputs_falcon_v2_topk5.json"

print("□ Phase 2 TOP_K:", TOP_K_V2)
print("□ Phase 2 Adapter dir:", ADAPTER_DIR_V2)
print("□ Phase 2 Output JSON:", FT_OUTPUT_PATH_V2)
```

```
□ Phase 2 TOP_K: 5
□ Phase 2 Adapter dir: /content/drive/MyDrive/Mapua /3rd Year Term
2/CSS181-3/NLP_Project/Models/falcon_finetune_adapter_v2_topk5
□ Phase 2 Output JSON: /content/drive/MyDrive/Mapua /3rd Year Term
2/CSS181-3/NLP_Project/Results/finetuned_outputs_falcon_v2_topk5.json
```

This section defines the experimental configuration for the refined fine-tuning phase. Retrieval depth is increased to TOP_K = 5 to provide broader contextual coverage. New directories are defined for saving Phase 2 adapters and evaluation outputs to ensure that results from Phase 1 remain preserved. This separation guarantees a controlled experimental comparison between baseline, Phase 1, and Phase 2 configurations without overwriting previous artifacts.

4.3 Retrieval Utilities

```
def retrieve_top_k_chunks_v2(query: str, top_k: int = TOP_K_V2):
    q_emb = embedder.encode([query],
convert_to_numpy=True).astype(np.float32)
    distances, indices = index.search(q_emb, top_k)
    idxs = indices[0].tolist()

    retrieved = []
    for rank, i in enumerate(idxs):
        row = meta[i]
        retrieved.append({
            "faiss_id": i,
            "text": row.get("text", ""),
            "source": row.get("source", ""),
            "page_number": row.get("page_number", None),
            "distance": float(distances[0][rank]),
        })
    return retrieved


def build_rag_input_v2(retrieved_chunks, question: str, top_k: int =
TOP_K_V2) -> str:
    context_blocks = []
    for c in retrieved_chunks:
        context_blocks.append(
```

```
            f"[Source: {c['source']} | Page: {c['page_number']}]\
n{c['text']}"
        )

    context = "\n\n---\n\n".join(context_blocks)

    return (
        "You are a legal assistant. Answer the question using ONLY the
retrieved context.\n"
        "If the context does not contain the answer, say: \"Not found
in the provided context.\"\n\n"
        f"Retrieved Context (Top-{top_k}):\n{context}\n\n"
        f"Question:\n{question}\n\n"
        "Answer:"
    )
```

To support the updated retrieval depth, retrieval and prompt-construction utilities are adapted to dynamically incorporate TOP_K = 5 context chunks. These functions maintain the same instruction template used in Phase 1, ensuring that the only structural change between experiments is retrieval coverage. By keeping the prompt format consistent, any observed performance differences can be attributed to retrieval depth and training refinement rather than prompt redesign.

4.4 Dataset Construction

```
from datasets import Dataset
from sentence_transformers import SentenceTransformer
import numpy as np

embed_model_name = "all-MiniLM-L6-v2"
embedder = SentenceTransformer(embed_model_name)

train_rows_v2 = []

for p in prompts:
    question = p.get("question", "")
    expected = join_expected_output(p.get("expected_output"))

    retrieved = retrieve_top_k_chunks_v2(question, top_k=TOP_K_V2)
    prompt_text = build_rag_input_v2(retrieved, question,
top_k=TOP_K_V2)

    train_rows_v2.append({
        "prompt_id": p.get("prompt_id"),
        "prompt_text": prompt_text,
        "target_text": expected
    })
```

```
ds_train_v2 = Dataset.from_list(train_rows_v2)

print("□ Phase 2 training samples:", len(ds_train_v2))
```

{"model_id":"ea503c7d530749709f9bc70510698ca5","version_major":2,"version_minor":0}

```
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different
task/architecture; not ok if you expect identical arch.

□ Phase 2 training samples: 10
```

In this step, the supervised fine-tuning dataset is reconstructed using the expanded retrieval configuration. For each prompt, the top five semantically similar legal chunks are retrieved from the FAISS index and embedded into the RAG instruction template. The expected output remains unchanged from Phase 1, preserving evaluation consistency. This reconstruction ensures that training supervision is better aligned with available contextual evidence.

4.5 Tokenization & Maskingt

```
from transformers import AutoTokenizer

MODEL_ID = "tiiuae/falcon-7b-instruct"

tokenizer = AutoTokenizer.from_pretrained(
    MODEL_ID,
    cache_dir=HF_CACHE_DIR
)

if tokenizer.pad_token is None:
    tokenizer.pad_token = tokenizer.eos_token

MAX_LEN_V2 = 1024

def tokenize_and_mask_v2(batch):
    prompt = batch["prompt_text"]
    target = batch["target_text"]

    full = prompt + " " + target
    tok_full = tokenizer(full, truncation=True, max_length=MAX_LEN_V2)

    tok_prompt = tokenizer(prompt, truncation=True,
max_length=MAX_LEN_V2)
```

```
    prompt_len = len(tok_prompt["input_ids"])

    labels = tok_full["input_ids"].copy()
    labels[:prompt_len] = [-100] * prompt_len
    tok_full["labels"] = labels
    return tok_full

ds_train_tok_v2 = ds_train_v2.map(
    tokenize_and_mask_v2,
    remove_columns=ds_train_v2.column_names
)

print("□ Phase 2 tokenized dataset ready.")
```

{"model_id":"389a376817744286a4f88cb3e933faa3","version_major":2,"version_minor":0}

```
□ Phase 2 tokenized dataset ready.
```

The dataset is tokenized using the same masking strategy applied in Phase 1. Prompt tokens are masked from loss computation to ensure that gradient updates are driven exclusively by the target answer text. This approach encourages the model to learn improved answer generation behavior rather than memorizing prompt structure. Maintaining identical masking methodology ensures fairness across experimental phases.

4.6 Model Reinitialization

```
import gc
import torch
from transformers import AutoModelForCausalLM, BitsAndBytesConfig

gc.collect()
torch.cuda.empty_cache()

bnb_config_v2 = BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_compute_dtype=torch.float16,
    bnb_4bit_use_double_quant=True,
    bnb_4bit_quant_type="nf4",
)

print("Loading base Falcon for Phase 2...")

base_model_v2 = AutoModelForCausalLM.from_pretrained(
    MODEL_ID,
    cache_dir=HF_CACHE_DIR,
    device_map={"": 0},
    quantization_config=bnb_config_v2,
```

```
    low_cpu_mem_usage=True
)

base_model_v2.train()
print("□ Base model loaded for Phase 2.")
```

```
Loading base Falcon for Phase 2...
```

{"model_id":"cb098df67c8e418a84a6b8a3ad053e6f","version_major":2,"version_minor":0}

```
The tied weights mapping and config for this model specifies to tie
transformer.word_embeddings.weight to lm_head.weight, but both are
present in the checkpoints, so we will NOT tie them. You should update
the config with `tie_word_embeddings=False` to silence this warning

□ Base model loaded for Phase 2.
```

To ensure experimental integrity, Phase 2 begins from a fresh base Falcon-7B model rather than continuing from the Phase 1 adapter. This prevents compounding adapter effects and guarantees that improvements (or regressions) can be attributed solely to the refined retrieval configuration and adjusted hyperparameters. The model is again loaded in 4-bit precision to maintain computational efficiency.

4.7 LoRA Configuration

```
from peft import LoraConfig, get_peft_model,
prepare_model_for_kbit_training

base_model_v2 = prepare_model_for_kbit_training(base_model_v2)

lora_config_v2 = LoraConfig(
    r=16,
    lora_alpha=32,
    lora_dropout=0.05,
    bias="none",
    task_type="CAUSAL_LM",
)

model_v2 = get_peft_model(base_model_v2, lora_config_v2)
model_v2.print_trainable_parameters()
```

```
trainable params: 4,718,592 || all params: 7,221,908,352 || trainable
%: 0.0653
```

The LoRA configuration remains consistent with Phase 1 in terms of rank, alpha, and dropout settings. By holding these parameters constant, the experiment isolates the effects of retrieval depth and learning rate adjustments. This controlled design supports a clearer interpretation of performance differences between phases.

4.8 Training Run

```
from transformers import TrainingArguments, Trainer,
DataCollatorForLanguageModeling

training_args_v2 = TrainingArguments(
    output_dir=f"{PROJECT_DIR}/Models/falcon_finetune_ckpts_v2_topk5",
    per_device_train_batch_size=1,
    gradient_accumulation_steps=8,
    num_train_epochs=5,
    learning_rate=2e-4,
    fp16=True,
    logging_steps=1,
    save_strategy="epoch",
    report_to="none"
)

data_collator_v2 = DataCollatorForLanguageModeling(
    tokenizer=tokenizer,
    mlm=False
)

trainer_v2 = Trainer(
    model=model_v2,
    args=training_args_v2,
    train_dataset=ds_train_tok_v2,
    data_collator=data_collator_v2
)

train_result_v2 = trainer_v2.train()
print(train_result_v2)
```

```
/usr/local/lib/python3.12/dist-packages/torch/_dynamo/
eval_frame.py:632: UserWarning: torch.utils.checkpoint: the
use_reentrant parameter should be passed explicitly. In version 2.5 we
will raise an exception if use_reentrant is not passed.
use_reentrant=False is recommended, but if you need to preserve the
current default behavior, you can pass use_reentrant=True. Refer to
docs for more details on the differences between the two variants.
  return fn(*args, **kwargs)
```

```
<IPython.core.display.HTML object>
```

```
/usr/local/lib/python3.12/dist-packages/torch/_dynamo/
eval_frame.py:632: UserWarning: torch.utils.checkpoint: the
use_reentrant parameter should be passed explicitly. In version 2.5 we
will raise an exception if use_reentrant is not passed.
use_reentrant=False is recommended, but if you need to preserve the
current default behavior, you can pass use_reentrant=True. Refer to
```

```
docs for more details on the differences between the two variants.
  return fn(*args, **kwargs)
/usr/local/lib/python3.12/dist-packages/torch/_dynamo/eval_frame.py:63
2: UserWarning: torch.utils.checkpoint: the use_reentrant parameter
should be passed explicitly. In version 2.5 we will raise an exception
if use_reentrant is not passed. use_reentrant=False is recommended,
but if you need to preserve the current default behavior, you can pass
use_reentrant=True. Refer to docs for more details on the differences
between the two variants.
  return fn(*args, **kwargs)
/usr/local/lib/python3.12/dist-packages/torch/_dynamo/eval_frame.py:63
2: UserWarning: torch.utils.checkpoint: the use_reentrant parameter
should be passed explicitly. In version 2.5 we will raise an exception
if use_reentrant is not passed. use_reentrant=False is recommended,
but if you need to preserve the current default behavior, you can pass
use_reentrant=True. Refer to docs for more details on the differences
between the two variants.
  return fn(*args, **kwargs)
/usr/local/lib/python3.12/dist-packages/torch/_dynamo/eval_frame.py:63
2: UserWarning: torch.utils.checkpoint: the use_reentrant parameter
should be passed explicitly. In version 2.5 we will raise an exception
if use_reentrant is not passed. use_reentrant=False is recommended,
but if you need to preserve the current default behavior, you can pass
use_reentrant=True. Refer to docs for more details on the differences
between the two variants.
  return fn(*args, **kwargs)

TrainOutput(global_step=10, training_loss=1.9241230368614197,
metrics={'train_runtime': 301.8976, 'train_samples_per_second': 0.166,
'train_steps_per_second': 0.033, 'total_flos': 2127802151731200.0,
'train_loss': 1.9241230368614197, 'epoch': 5.0})
```

In Phase 2, fine-tuning is executed using the reconstructed Top-5 retrieval dataset and the same QLoRA-based parameter-efficient training procedure applied in Phase 1. The training configuration is kept consistent to preserve experimental control, including the same batch size, gradient accumulation strategy, number of epochs, and learning rate. This ensures that the primary experimental difference remains the retrieval depth used to construct the contextual evidence for supervision.

4.9 Adapter Export

```
model_v2.save_pretrained(ADAPTER_DIR_V2)
tokenizer.save_pretrained(ADAPTER_DIR_V2)

print("□ Phase 2 adapter saved to:", ADAPTER_DIR_V2)

□ Phase 2 adapter saved to: /content/drive/MyDrive/Mapua /3rd Year
Term 2/CSS181-3/NLP_Project/Models/falcon_finetune_adapter_v2_topk5
```

Upon completion of training, the refined LoRA adapter is saved to a distinct directory. This ensures reproducibility and enables independent evaluation of Phase 2 results without interfering with prior experiments. Maintaining separate adapter artifacts supports clean comparative analysis.

4.10 Evaluation & JSON Export

```
from tqdm import tqdm

model_v2.eval()

@torch.inference_mode()
def generate_answer_v2(input_text: str, max_new_tokens: int = 256) ->
str:
    inputs = tokenizer(input_text,
return_tensors="pt").to(model_v2.device)
    out_ids = model_v2.generate(
        **inputs,
        do_sample=False,
        temperature=0.0,
        top_p=1.0,
        max_new_tokens=max_new_tokens,
        pad_token_id=tokenizer.eos_token_id
    )
    decoded = tokenizer.decode(out_ids[0], skip_special_tokens=True)
    return decoded[len(input_text):].strip()

ft_results_v2 = []

for p in tqdm(prompts, desc="Running Falcon Phase 2 (TopK=5)"):
    q = p["question"]
    retrieved = retrieve_top_k_chunks_v2(q, top_k=TOP_K_V2)
    input_text = build_rag_input_v2(retrieved, q, top_k=TOP_K_V2)
    answer = generate_answer_v2(input_text)

    ft_results_v2.append({
        "model_id": MODEL_ID,
        "adapter_dir": ADAPTER_DIR_V2,
        "prompt_id": p.get("prompt_id"),
        "method": p.get("method"),
        "condition": p.get("condition"),
        "dataset_file": p.get("dataset_file"),
        "section_index": p.get("section_index"),
        "question": q,
        "retrieved_chunks": retrieved,
        "answer": answer,
        "expected_output": p.get("expected_output"),
        "phase": "Phase 2 (TopK=5)"
```

```
    })

import json
with open(FT_OUTPUT_PATH_V2, "w", encoding="utf-8") as f:
    json.dump(ft_results_v2, f, indent=2, ensure_ascii=False)

print("□ Phase 2 outputs saved.")
```

```
 Running Falcon Phase 2 (TopK=5):   0%|          | 0/10 [00:00<?, ?
it/s]The following generation flags are not valid and may be ignored:
['temperature']. Set `TRANSFORMERS_VERBOSITY=info` for more details.
This is a friendly reminder - the current text generation call has
exceeded the model's predefined maximum length (2048). Depending on
the model, you may observe exceptions, performance degradation, or
nothing at all.
Running Falcon Phase 2 (TopK=5):  10%|▉         | 1/10 [00:25<03:50,
25.63s/it]Token indices sequence length is longer than the specified
maximum sequence length for this model (2232 > 2048). Running this
sequence through the model will result in indexing errors
Running Falcon Phase 2 (TopK=5): 100%|██████████| 10/10 [04:26<00:00,
26.60s/it]

□ Phase 2 outputs saved.
```

The fine-tuned Phase 2 model is evaluated using deterministic generation settings identical to those used in Phase 1 and the baseline evaluation. Responses are generated for the same set of prompts and saved to a new JSON file. This guarantees methodological consistency and enables direct quantitative comparison across all three experimental configurations.

4.11 Output Preview

```
print(f"\n□ Loaded {len(ft_results_v2)} Phase 2 fine-tuned outputs\n")

for item in ft_results_v2:
    print("=" * 80)
    print(f"[Prompt {item['prompt_id']}] ({item['method']})")
    print("-" * 80)
    print("Q:", item["question"])
    print("\nA:", item["answer"][:400])
    print("\n")
```

```
□ Loaded 10 Phase 2 fine-tuned outputs

================================================================
==========
[Prompt 1] (Zero-Shot)
----------------------------------------------------------------
```

```
----------
Q: List all the conditions under which the processing of personal
information is considered lawful according to this section.

A: The processing of personal information is considered lawful
according to this section if:

1. The data subject has given his or her consent.
2. The processing is necessary and is related to the fulfillment of a
contract with the data subject or in order to take steps at the
request of the data subject prior to entering into a contract.
3. The processing is necessary for compliance with a legal obl


==================================================================
==========
[Prompt 2] (Zero-Shot)
------------------------------------------------------------------
----------
Q: What are the penalties for unauthorized disclosure of personal
information and sensitive personal information under this section?

A: The penalties for unauthorized disclosure of personal information
and sensitive personal information under this section are:

1. For personal information, the penalty is imprisonment for 1 year to
3 years and a fine of not less than 5 million pesos (php5,000,000.00)
but not more than 2 million pesos (php2,000,000.00)

2. For sensitive personal information, the penalty is imprisonment for
1 year to


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[Prompt 3] (Chain-of-Thought Prompting)
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Q: Based on the legal definition, list what qualifies as sensitive
personal information and briefly explain why each qualifies.

A: 1. Personal information about an individual that is or was an
officer or employee of government that relates to: a. an individual's
personal information, including but not limited to, name, birthdate,
address, phone number, email, and other personal information; b. an
individual's personal information, including but not limited to, name,
birthdate, address, phone number, email, and other personal


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[Prompt 4] (Chain-of-Thought Prompting)

Q: Explain the general data privacy principles outlined in this section and why each principle is important.

A: 1. Consent. Consent is the cornerstone of data privacy. Consent is required for the collection, processing, and retention of personal data. Consent must be given by the data subject, and the data subject must be informed of the purpose of the collection, processing, and retention of personal data. Consent must be given by the data subject, and the data subject must be informed of the purpose of th

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[Prompt 5] (One-Shot Learning Prompt)

Q: List the rights granted to data subjects under this section.

A: The rights granted to data subjects under this section are:

1. Right to be informed
2. Right to access
3. Right to rectification
4. Right to erasure
5. Right to data portability
6. Right to object
7. Right to restriction of processing
8. Right to data portability
9. Right to withdraw consent
10. Right to complain to the data protection authority
11. Right to access personal data
12. Right to acce

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[Prompt 6] (One-Shot Learning Prompt)

Q: Summarize the breach notification requirement.

A: The breach notification requirement is that a personal information controller must notify the Commission of any breach of personal data within 72 hours of the breach. The notification must include the following:

1. The name of the personal information controller.
2. The name of the person responsible for the breach.
3. The date of the breach.
4. The type of personal data that was breached.
5. The

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[Prompt 7] (One-Shot Learning Prompt)

Q: Explain what the principle of proportionality means.

A: The principle of proportionality is a legal principle that means that the severity of the legal measure must be proportionate to the nature and extent of the harm caused by the act or omission. In other words, the severity of the legal measure must be in proportion to the harm caused by the act or omission. The principle of proportionality is a legal principle of the rule of law. It is a legal pri

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[Prompt 8] (One-Shot Learning Prompt)

Q: List the three types of security measures required.

A: 1. Physical security measures:
- CCTV cameras
- Security gates
- Security barriers
- Motion detectors
- Fire alarms
- CCTV systems
- CCTV systems
- Security systems
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[Prompt 9] (Negative Prompting)
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Q: List only the lawful processing conditions mentioned.

A: 1. Consent of the data subject
2. Processing is necessary to achieve the lawful and noncommercial
objectives of public organizations and their associations
3. Processing is necessary to achieve the lawful and noncommercial
objectives of public organizations and their associations
4. Processing is necessary to achieve the lawful and noncommercial
objectives of public organizations and their associa


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[Prompt 10] (Negative Prompting)
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Q: Identify the penalties for unauthorized disclosure.

A: The penalties for unauthorized disclosure of personal information
and sensitive personal information are:

1. Unauthorized processing of personal information:
- A penalty of up to P1 million or imprisonment for up to 6 years, or
both.
- A penalty of up to P1 million or imprisonment for up to 6 years, or
both.

2. Unauthorized processing of sensitive personal information:
- A penalty of up to P1 mi
```

This section provides a preview of the Phase 2 generated responses after evaluation and JSON export. Displaying the outputs directly in the notebook allows quick verification that the refined Top-5 configuration produced valid responses for each prompt before formal comparison in the succeeding section.

Section 5: Phase 1 vs Phase 2 Result Comparison

5.1 Load Saved Experimental Outputs

```
PHASE1_RESULTS_PATH =
f"{PROJECT_DIR}/Results/finetuned_outputs_falcon_phase1_original.json"
PHASE2_RESULTS_PATH =
f"{PROJECT_DIR}/Results/finetuned_outputs_falcon_v2_topk5.json"

with open(PHASE1_RESULTS_PATH, "r", encoding="utf-8") as f:
    phase1_results = json.load(f)

with open(PHASE2_RESULTS_PATH, "r", encoding="utf-8") as f:
    phase2_results = json.load(f)

print("□ Phase 1 outputs loaded:", len(phase1_results))
print("□ Phase 2 outputs loaded:", len(phase2_results))
```

```
□ Phase 1 outputs loaded: 10
□ Phase 2 outputs loaded: 10
```

This subsection reloads the saved JSON outputs from Phase 1 and Phase 2 so that both experimental configurations can be reviewed within the same notebook session. Using the exported files also helps preserve reproducibility by ensuring that the displayed outputs correspond to the saved experimental artifacts.

5.2 Side-by-Side Comparison

```
phase1_by_id = {item["prompt_id"]: item for item in phase1_results}
phase2_by_id = {item["prompt_id"]: item for item in phase2_results}

all_prompt_ids = sorted(set(phase1_by_id.keys()) |
set(phase2_by_id.keys()))

for pid in all_prompt_ids:

    p1 = phase1_by_id.get(pid)
    p2 = phase2_by_id.get(pid)

    question = None
    expected = None

    if p1:
```

```
        question = p1.get("question")
        expected = p1.get("expected_output")
    elif p2:
        question = p2.get("question")
        expected = p2.get("expected_output")

    print("=" * 100)
    print(f"Prompt ID: {pid}")

    print("\nQuestion:")
    print(question)

    print("\nExpected:")
    print(expected)

    if p1:
        print("\n[Phase 1]")
        print(p1["answer"][:600])

    if p2:
        print("\n[Phase 2]")
        print(p2["answer"][:600])

    print("\n")
```

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Prompt ID: 1

Question:
List all the conditions under which the processing of personal information is considered lawful according to this section.

Expected:
['Consent of the data subject', 'Fulfilling contractual obligations', 'Compliance with legal obligations', 'Protection of vitally important interests', 'Responding to national emergency or public order', 'Fulfillment of statutory mandate of a public authority', 'Legitimate interests of controller or third parties, except when overridden by fundamental rights']

[Phase 1]
The processing of personal information is considered lawful according to this section when:

1. The personal information is collected from the data subject.
2. The personal information is collected for a specific purpose.
3. The personal information is collected from a public authority or a government entity.
4. The personal information is collected from a public body or a
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government entity.
5. The personal information is collected from a public body or a government entity.
6. The personal information is collected from a public body or a government entity.
7. The personal information is collec

[Phase 2]
The processing of personal information is considered lawful according to this section if:

1. The data subject has given his or her consent.
2. The processing is necessary and is related to the fulfillment of a contract with the data subject or in order to take steps at the request of the data subject prior to entering into a contract.
3. The processing is necessary for compliance with a legal obligation to which the personal information controller is subject.
4. The processing is necessary for the establishment, exercise or defense of a legal claim.
5. The processing is necessary for the prot

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Prompt ID: 2

Question:
What are the penalties for unauthorized disclosure of personal information and sensitive personal information under this section?

Expected:
['Unauthorized disclosure of personal information: 1–3 years imprisonment, Php500,000–1,000,000 fine', 'Unauthorized disclosure of sensitive personal information: 3–5 years imprisonment, Php500,000–2,000,000 fine']

[Phase 1]
The penalties for unauthorized disclosure of personal information and sensitive personal information under this section are imprisonment ranging from one (1) year to three (3) years and a fine of not less than five hundred thousand pesos (php500,000.00) but not more than one million pesos (php1,000,000.00).

[Phase 2]
The penalties for unauthorized disclosure of personal information and sensitive personal information under this section are:

1. For personal information, the penalty is imprisonment for 1 year to 3 years and a fine of not less than 5 million pesos (php5,000,000.00) but not more than 2 million pesos (php2,000,000.00)

2. For sensitive personal information, the penalty is imprisonment for 1 year to 3 years and a fine of not less than 5 million pesos (php5,000,000.00) but not more than 2 million pesos (php2,000,000.00)

3. For both personal information and sensitive personal information, the pen

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Prompt ID: 3

Question:
Based on the legal definition, list what qualifies as sensitive personal information and briefly explain why each qualifies.

Expected:
['Race, ethnic origin, marital status, age, color, religious, philosophical or political affiliations, health, education, genetic or sexual life', 'Proceedings for any offense committed or alleged', 'Government-issued identifiers (SSS, GSIS, TIN, etc.)', 'Classified information by executive order or law', 'Bank account and financial information']

[Phase 1]
1. Race, ethnic origin, marital status, age, color, and religious, philosophical or political affiliations: These are personal information that are considered sensitive because they are often used to identify a person and can be used to discriminate against them.

2. Health, education, genetic or sexual life of a person: These are personal information that are considered sensitive because they can be used to identify a person and can be used to discriminate against them.

3. Social security numbers, previous or current health records, licenses or its denials, suspension or revocation, and tax

[Phase 2]
1. Personal information about an individual that is or was an officer or employee of government that relates to: a. an individual's personal information, including but not limited to, name, birthdate, address, phone number, email, and other personal information; b. an individual's personal information, including but not limited to, name, birthdate, address, phone number, email, and other personal information; c. an individual's personal information, including but not limited to, name, birthdate, address, phone number, email, and other personal information; d. an individual's personal informati

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Prompt ID: 4

Question:
Explain the general data privacy principles outlined in this section and why each principle is important.

Expected:
['Purpose limitation', 'Transparency and fairness', 'Accuracy', 'Storage limitation', 'Security safeguards']

[Phase 1]
The general data privacy principles outlined in this section are:

1. Data privacy and security: This principle ensures that personal data is collected, processed, and stored securely and in accordance with the data subject's rights. This is important because personal data is sensitive and confidential, and should be protected to prevent unauthorized access, use, or disclosure.

2. Consent: This principle ensures that personal data is collected, processed, and stored only with the consent of the data subject. This is important because personal data is sensitive and confidential, and should onl

[Phase 2]
1. Consent. Consent is the cornerstone of data privacy. Consent is required for the collection, processing, and retention of personal data. Consent must be given by the data subject, and the data subject must be informed of the purpose of the collection, processing, and retention of personal data. Consent must be given by the data subject, and the data subject must be informed of the purpose of the collection, processing, and retention of personal data. Consent must be given by the data subject, and the data subject must be informed of the purpose of the collection, processing, and retention o

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Prompt ID: 5

Question:
List the rights granted to data subjects under this section.

Expected:
['Right to be informed', 'Right to access', 'Right to object', 'Right to erasure or blocking', 'Right to damages', 'Right to data portability', 'Right to lodge a complaint']

[Phase 1]
The rights granted to data subjects under this section are:

1. Right to access personal data
2. Right to data portability
3. Right to be informed
4. Right to rectification
5. Right to erasure
6. Right to restriction of processing
7. Right to object to processing
8. Right to data portability
9. Right to withdraw consent
10. Right to be forgotten
11. Right to be informed of a data breach
12. Right to access a copy of the data
13. Right to access a copy of the data in a machine-readable format
14. Right to access a copy of the data in a machine-readable format
15. Right to access a copy of the d

[Phase 2]
The rights granted to data subjects under this section are:

1. Right to be informed
2. Right to access
3. Right to rectification
4. Right to erasure
5. Right to data portability
6. Right to object
7. Right to restriction of processing
8. Right to data portability
9. Right to withdraw consent
10. Right to complain to the data protection authority
11. Right to access personal data
12. Right to access personal data
13. Right to access personal data
14. Right to access personal data
15. Right to access personal data
16. Right to access personal data
17. Right to access personal data
18. Right to

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Prompt ID: 6

Question:
Summarize the breach notification requirement.

Expected:
['Notify the National Privacy Commission and affected data subjects within 72 hours if the breach risks rights and freedoms']

[Phase 1]
The breach notification requirement is that a personal information controller must notify the National Privacy Commission (NPC) of any security breach or personal data breach that may have occurred. The notification must include the type of breach, the personal data that was affected, the date of the breach, and the contact information of the person responsible for the breach. The notification must also include measures taken to address the breach and any assistance to be provided to the affected data subjects.

[Phase 2]
The breach notification requirement is that a personal information controller must notify the Commission of any breach of personal data within 72 hours of the breach. The notification must include the following:

1. The name of the personal information controller.
2. The name of the person responsible for the breach.
3. The date of the breach.
4. The type of personal data that was breached.
5. The number of data subjects affected by the breach.
6. The measures taken by the personal information controller to prevent further breaches.
7. The date of the last update of the personal data.
8. The d

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Prompt ID: 7

Question:
Explain what the principle of proportionality means.

Expected:
['Personal data collected must be adequate, relevant, necessary, and not excessive in relation to the declared purpose']

[Phase 1]
The principle of proportionality means that the amount of personal data collected and processed should be in proportion to the purpose for which it is being collected and processed. In other words, the amount of personal data collected and processed should be in proportion to the legitimate purpose of the collection and processing.

[Phase 2]

The principle of proportionality is a legal principle that means that the severity of the legal measure must be proportionate to the nature and extent of the harm caused by the act or omission. In other words, the severity of the legal measure must be in proportion to the harm caused by the act or omission. The principle of proportionality is a legal principle of the rule of law. It is a legal principle of the rule of law that is found in the United States Constitution. The principle of proportionality is a legal principle of the rule of law that is found in the United States Constitution. The

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Prompt ID: 8

Question:
List the three types of security measures required.

Expected:
['Organizational measures', 'Physical measures', 'Technical measures']

[Phase 1]
1. Physical security measures
2. Data security measures
3. Network security measures

[Phase 2]
1. Physical security measures:
- CCTV cameras
- Security gates
- Security barriers
- Motion detectors
- Fire alarms
- CCTV systems
- CCTV systems
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Prompt ID: 9

Question:
List only the lawful processing conditions mentioned.

Expected:
['Consent', 'Contractual necessity', 'Legal obligation', 'Vital interests', 'Public order', 'Statutory mandate', 'Legitimate interests not overridden by fundamental rights']

[Phase 1]
1. Consent of the data subject
2. Processing is necessary for the performance of a contract
3. Processing is necessary for the purpose of a legitimate interest pursued by the controller or a third party
4. Processing is necessary for the purpose of a public interest
5. Processing is necessary for the purpose of a public authority
6. Processing is necessary for the purpose of a public interest
7. Processing is necessary for the purpose of a public interest
8. Processing is necessary for the purpose of a public authority
9. Processing is necessary for the purpose of a public interest
10. Process

[Phase 2]
1. Consent of the data subject
2. Processing is necessary to achieve the lawful and noncommercial objectives of public organizations and their associations
3. Processing is necessary to achieve the lawful and noncommercial objectives of public organizations and their associations
4. Processing is necessary to achieve the lawful and noncommercial objectives of public organizations and their associations
5. Processing is necessary to achieve the lawful and noncommercial objectives of public organizations and their associations
6. Processing is necessary to achieve the lawful and noncommercial ob

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Prompt ID: 10

Question:
Identify the penalties for unauthorized disclosure.

Expected:
['Personal information: 1–3 years imprisonment, Php500,000–1,000,000 fine', 'Sensitive personal information: 3–5 years imprisonment, Php500,000–2,000,000 fine']

[Phase 1]
The penalties for unauthorized disclosure are imprisonment ranging from one (1) year to three (3) years and a fine of not less than five hundred thousand pesos (php500,000.00) but not more than one million pesos (php1,000,000.00).

[Phase 2]
The penalties for unauthorized disclosure of personal information and sensitive personal information are:

1. Unauthorized processing of personal information:
- A penalty of up to P1 million or imprisonment for up to 6 years, or both.
- A penalty of up to P1 million or imprisonment for up to 6 years, or both.

2. Unauthorized processing of sensitive personal information:
- A penalty of up to P1 million or imprisonment for up to 6 years, or both.
- A penalty of up to P1 million or imprisonment for up to 6 years, or both.

3. Unauthorized processing of personal information and sensitive personal